

A Prospective Comparative Study of Circulating Tumour DNA Analysis for Detecting EGFR Mutations in Non-Small Cell Lung Carcinoma Patients Using BEAMing PCR

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Abstract

Background: Somatic DNA mutations are highly tumor specific which can be used as optimum diagnostic markers. Noninvasive mutation detection by assessing circulating tumor DNA (ctDNA) offers many advantages over tumor biopsy.

Methods and findings: ctDNA from plasma of 100 Non-Small Cell Lung Cancer (NSCLC) patients were analyzed for the presence of EGFR activating mutations using highly sensitive BEAMing Polymerase Chain Reaction (PCR). We could identify EGFR exon 19, 20 and 21 exon mutations in ctDNA. These data were compared with the qPCR data available in our hospital patient Electronic Medical Records (EMR). McNemar's test was done to determine the concordance between BEAMing and qPCR done on primary tumor tissues. Our results showed high concordance with *Chi square* p value of 1.00 for all the exons (19, 20 and 21) between BEAMing and EMR-qPCR.

Keywords: EGFR; NSCLC; ctDNA; BEAMing; qPCR

Abbreviations: EGFR: Epidermal growth factor receptor; BEAMing: Beads Emulsion Amplification Magnetics

Introduction

Despite several advances in therapy, lung cancer remains the leading cause of cancer related death worldwide with only 15% patients survive beyond 5 years [1]. Among the Non-Small Cell Lung Carcinoma (NSCLC), adenocarcinoma constitutes the most common histologic subtype [2]. In most of the cases lung cancer patients present with locally advanced or metastatic disease and are not fit for surgical resection. Epidermal Growth Factor Receptor (EGFR) gene is located on the short arm of chromosome 7 (7p 11.2) and encodes a 170 KDa type I transmembrane growth factor receptor with Tyrosine Kinase (TK) activity [3]. EGFR belongs to the HER/erbB family of Receptor Tyrosine Kinases (RTKs), which includes HER1 (EGFR/erbB1), HER2 (neu, erbB2), HER3 (erbB3) and HER4 (erbB4). They have an extracellular, cysteine-rich ligand binding domain; a single α -helix transmembrane domain; a cytoplasmic TK domain (in all receptors except HER3) and a carboxy terminal signaling domain [4]. The TK activity of EGFR may be dysregulated by several oncogenic mechanisms, including EGFR gene mutation, increased gene copy number, and EGFR protein overexpression [5]. There are two types of mutations reported mostly in the TK domain of EGFR [6,7] short deletions in exon 19 and point mutations (G719S, L858R, and L861Q) in exons 19, 20 and 21. In its inactive form, the EGFR kinase domain assumes a structure that prevents its activation [8]. These mutations result in the destabilization of its domain conformation, constitutive activation of its kinase activity, and activation of its downstream signaling pathways [9,10] like AKT and STAT, which have crucial anti-apoptosis functions for cell survival [11].

In cancer patients, elevated levels of nucleic acids are released into the blood circulation. As tumor grows, some of its cells dies and releases its DNA into the circulation [12]. The amount of ctDNA itself can be used as a marker, as tumor cells release more DNA than healthy cells. This leads to the development of sensitive assays to quantify tumor load and detect relapses and hidden metastases [13].

Recently, circulating tumor DNA (ctDNA) has emerged as a specific and sensitive blood-based biomarker for detection of different mutations including EGFR.

Most of the NSCLC patients with EGFR mutations, though initially respond to EGFR-Tyrosine Kinase Inhibitors (TKI), develop drug resistance later. This attributes to the EGFR T70M mutations to first-line EGFR-TKIs [14,15]. Earlier studies have demonstrated that ctDNA detection of *EGFR* mutations including T790M are highly concordant with those detected in tumor tissues [16-20].

In the present study, we used bead based PCR assay (beads, emulsion, analytics and magnetics, BEAMing) to detect different activating (exon 19, 21) and resistant *EGFR* mutations (exon 20) in ctDNA derived from the peripheral blood of the lung cancer

patients. We compare these results with the standard qPCR as well as with a standardized kit. Results suggest that ctDNA may complement the biopsy of primary lesions as a source of *EGFR* mutation detection.

Materials and Methods

Patient samples and cell lines

This study was approved by the Institutional Review Board of ACTREC-Tata Memorial Centre, India. Patients with activating EGFR mutations in tumor tissues were selected following a biopsy examination between June 2017 and April 2020 (Table 1). Peripheral blood was collected from 100 patients with lung cancer treated at Tata Memorial Centre-ACTREC, India. Research was conducted following Good Clinical Practice. All patients were provided with written informed consent prior to participation. Total genomic DNA isolated from PC9, H1975 and A549 NSCLC cell lines (kind gift from Dr. Amit Dutt, ACTREC) were used as controls. Formalin-fixed paraffin embedded tumor tissue was obtained from each subject and processed by the Surgical Pathology Laboratory using routine procedures.

Table 1: Patient characteristics.

Characteristics	Number (%)
Age	
Mean	53.07
Median	54
Sex	
Male	67 (67%)
Female	33 (33%)
Pathological diagnosis	
Adenocarcinoma	45
Metastatic adenocarcinoma	42
Metastatic squamous carcinoma	2
Mucoepidermoid carcinoma	1
Squamous carcinoma	8
Hepatoid carcinoma	1
ND	1

Isolation of circulating tumor DNA (ctDNA)

Blood was drawn into EDTA tubes (for plasma). Each time 10 ml blood was collected. Within one hour, the tubes were subjected to centrifugation at 820 g for 10 min. 1 ml aliquots of the plasma was transferred to 1.5 ml tubes and centrifuged at

16,000 g for 10 min to pellet any remaining cellular debris. Supernatants were transferred to fresh tubes and stored at -80°C. Total genomic DNA was extracted from 1 ml plasma using DNA Micro kit (Qiagen) according to the manufacturer's instructions.

Quantification of total plasma DNA

The isolated ctDNA concentration was estimated using Nanodrop ND1000 spectrophotometer.

Sensitivity and specificity assay

Sensitivity and specificity of the BEAMing reaction was done before starting the actual assay. In this, different percentages of mutant DNAs from cell lines harbouring specific mutations were mixed in the background of wild-type DNA. Emulsion based BEAMing PCR was done and hybridized samples were run in FACSArea to detect mutant and wild type populations.

BEAMing and mutation detection

Different primer sets were designed for the mutation analysis Table 2 [21,22]. The DNA purified from the plasma was used for each BEAMing assay. Initially an amplification with a high-fidelity DNA polymerase was performed in eight separate 25 µl PCR reactions each containing template DNA from 250 µl of plasma, 5× Phusion High Fidelity PCR buffer (NEB), 1.5U of Hotstart Phusion polymerase (NEB), 0.2 µM of each primer, 0.25 mM of each dNTP, and 0.5 mM MgCl₂. Temperature cycling was: 98°C 30s, 35 × (98°C 10 s, 57°C 10 s, 72°C 10 s). PCR products were pooled and quantified using the nanodrop spectrophotometer.

Table 2: Primer sequences.

Exon	Name	Sequence	Modification	Target
Primers for Exon amplification				
Exon 19	Tag1-19del	TCCCGCGAAATTAATA CGACAAGTTAAAATTC CCGTCGCTATC		
	Tag2-19del	GCTGGAGCTCTGCAG CTAGACCCCCACACAG CAAAG		
Exon 20	Tag1-T790M	TCCCGCGAAATTAATA CGACGCATCTGCCTCA CCTCCAC		
	Tag2-T790M	GCTGGAGCTCTGCAG CTAAGCAGGTACTGGG AGCCAAT		
Exon 21	Tag1-L858R	TCCCGCGAAATTAATA CGACAGCCAGGAACG TACTGGTGA		
	Tag2-L858R	GCTGGAGCTCTGCAG CTATGCCTCCTTCTGC ATGGTAT		
Emulsion PCR primers				
	Tag1-F	TCCCGCGAAATTAATA CGAC		
	Tag2-R	GCTGGAGCTCTGCAG CTA		
Hybridization probe for BEAMing				
Exon 19	19del_35_49_Cy5	GGAGATGTTTTGATAG CG	5' Cy5	exon 19 E746-A750del
	19del_36_50_Cy5	CGGAGATGTCTTGATA GC	5' Cy5	exon 19 E746-A750del
	19del_40_57 Cy5	TGGCTTTCGATTCCTT GA	5' Cy5	exon 19 L747-S752del.P753S

	19del_AATTCC_Cy5	TGTTGCTTCTCTTGGATT	5' Cy5	exon 19 E746-L747del.IP
	19del_36_55_T_Cy5	GCTTTCGGAACCTTGTAG	5' Cy5	exon 19 L747-S752del.E746V
	19del_35_53_ACT_Cy5	GGAGAAGTTTTGATAGCG	5' Cy5	exon 19 K745-E749del.A750K
	19del_39_56_CAG_Cy5	TTTCGGCTGTTCTTGAT	5' Cy5	exon 19 L747-T751del.S752Q
	19del_39_48_C_Cy5	GAGATGTTGGTTCCTTGAT	5' Cy5	exon 19 L747-E749del.A750P
	19del_WT_488	TGTTGCTTCTCTTAATTCC	5' Alexa488	exon 19 wild-type control
Exon 20	T790M_Mut- Cy5	atgagctgcAtgatgag	5' Cy5	T790M mutation
Exon 21	T790M_WT-488	tgagctgcGtgatgag	5' Alexa488	Wild-type control for T790M
	L858R_Mut_LNA_Cy5	gtttggccCgccccaaat	5' Cy5	L858R mutation
	L858R_WT_LNA_488	gtttggccAgccccaaat	5' Alexa488	Wild-type control for L858R
	L861Q_Mut_Cy5	caccagcTgtttggcc	5' Cy5	L861Q mutation
	L861Q_WT_488	caccagcAgttttggcc	5' Alexa488	Wild-type control for L861Q

Emulsion PCR was performed as follows [21]. Briefly, a 150 µl PCR mixture was prepared containing 18 pg template DNA, 40U of Platinum Taq DNA polymerase (Invitrogen), 1 x PCR buffer, 0.2 mM dNTPs, 5 mM MgCl₂, 0.05 µM Tag1 (5'-tcccgcgaaattaatacgac-3'), 8 µM Tag2 (5'-gctggagctctgcagcta-3') and ~6 × 10⁷ magnetic streptavidin beads (MyOne, Invitrogen) coated with Tag1 oligonucleotide (5'-dual biotin-T-Spacer18-tcccgcgaaattaatacgac-3'). The 150 µl PCR reaction, 600 µl oil/emulsifier mix (7% ABIL WE09, 20% mineral oil, 73% Tegosoft DEC (Evonik) and one 5 mm steel bead (Qiagen) were added to a 96 deep well plate 1.2 ml (Abgene). Emulsions were prepared by shaking the plate in a TissueLyser (Qiagen) for 10 s at 15 Hz and then 7 s at 17 Hz. Emulsions were dispensed into eight PCR wells and temperature cycled at 94°C for 2 min; 3 cycles of 94°C for 10 s, 68°C for 45 s, 70°C for 75s; 3 cycles of 94°C for 10 s, 65°C for 45 s, 70°C for 75 s, 3 cycles of 94°C for 10 s, 62°C for 45 s, 70°C for 75 s; 50 cycles of 94°C for 10 s, 57°C for 45 s, 70°C for 75 s. To break the emulsions, 150 µl breaking buffer (10 mM Tris-HCl, pH 7.5, 1% Triton-X 100, 1% SDS, 100 mM NaCl, 1 mM EDTA) was added to each well and mixed with a TissueLyser at 20Hz for 20 s. Beads were recovered by spinning the suspension at 3,200 g for 2 min and removing the oil phase. The breaking step was repeated twice. All beads from 8 wells were consolidated and washed with 150 µl wash buffer (20 mM Tris-HCl, pH 8.4, 50 mM KCl). The DNA on the beads was denatured for 5 min with 0.1 M NaOH. Finally, beads were washed with 150µl wash buffer and resuspended in 150 µl of the same buffer.

The mutation status of DNA bound to beads was determined by allele-specific hybridization. Fluorescently labeled probes complementary to the mutant and wild-type DNA sequences were designed for different mutations (Table 2). The size of the probes ranged from 15 to 18 nt. All mutant probes were coupled to a Cy5 fluorophore and all wild-type probes were coupled to a Alexa488 fluorophore at their 5' ends (Integrated DNA Technologies). Each allele-specific hybridization reaction contained ~1 × 10⁷ beads in 30 µl wash buffer, 66 µl of 1.5 × hybridization buffer (4.5 M tetramethylammonium chloride, 75 mM Tris-HCl pH 7.5, 6 mM EDTA), and 4 µl of a mixture of mutant and wild-type probes, each at 5 µM in TE buffer. The hybridization mixture was heated to 70°C for 10 s and slowly (0.1°C/s) cooled to 35°C. After incubating at 35°C for 2 min, the mixture was cooled (0.1°C/sec) to room temperature. The beads were collected with a magnet and the supernatant containing the unbound probes was removed using a pipette. The beads were resuspended in 100 µl of 1 × hybridization buffer and heated to 48°C for 5 min to remove unbound probes. After the heating step, beads were again separated magnetically and washed once with 100 µl wash buffer. In the final step, the supernatant was removed and beads were resuspended in 200 µl TE buffer for flow cytometric analysis. FACS Aria III flow cytometry system (BD Bioscience) equipped with a high throughput autosampler was used for the analysis of each bead population. An average of 1 × 10⁶ beads were analyzed for each plasma sample.

Real time PCR from Formalin Fixed Para in Embedded (FFPE tissue DNA: Paraffin blocks were collected from the Department of Pathology, Tata Memorial Hospital. A total of 50-100 μ m sections were used for the extraction of DNA. The sections were deparaffinized and stained with hematoxylin and eosin. All specimens had undergone histological examination to confirm the presence of tumor tissue. The dissected tumor tissues were digested overnight at 60°C in 180 μ l ATL buffer (Qiagen) and 20 μ l Proteinase K (50 mg/ml; Invitrogen). DNA was isolated using the QIAamp FFPE DNA Kit (Qiagen) following the manufacturer's protocol. The median quantity of extracted FFPE-DNA was estimated using a Nanodrop spectrophotometer. The presence of EGFR mutations was determined by real-time quantitative PCR (RQ-PCR) based-approach using Therascreen EGFR Mutation Test kit (Qiagen) as well as with Easy Line EGFR PCR kit (Diatech). Both the kits are designed to detect the most commonly reported *EGFR* mutations.

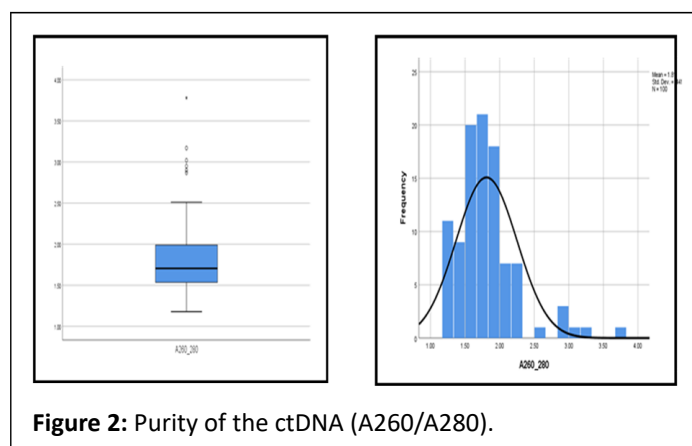
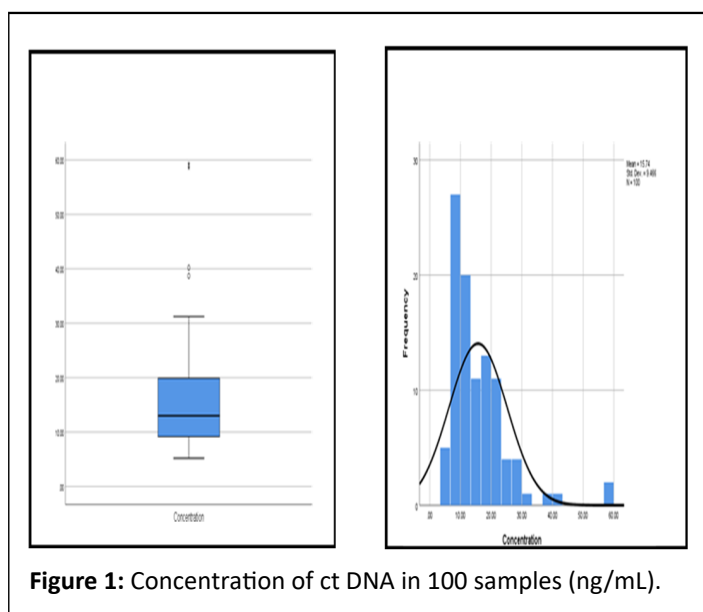
Statistical analysis

Statistical analysis was performed using IBM SPSS Windows version 20.0 software. Diagnostic measures such as sensitivity, specificity and accuracy were calculated. McNemar's test was used to test the statistical significance of the difference. Difference were compared between BEAMing data and EMR qPCR data as well as with Easy line qPCR data.

Results

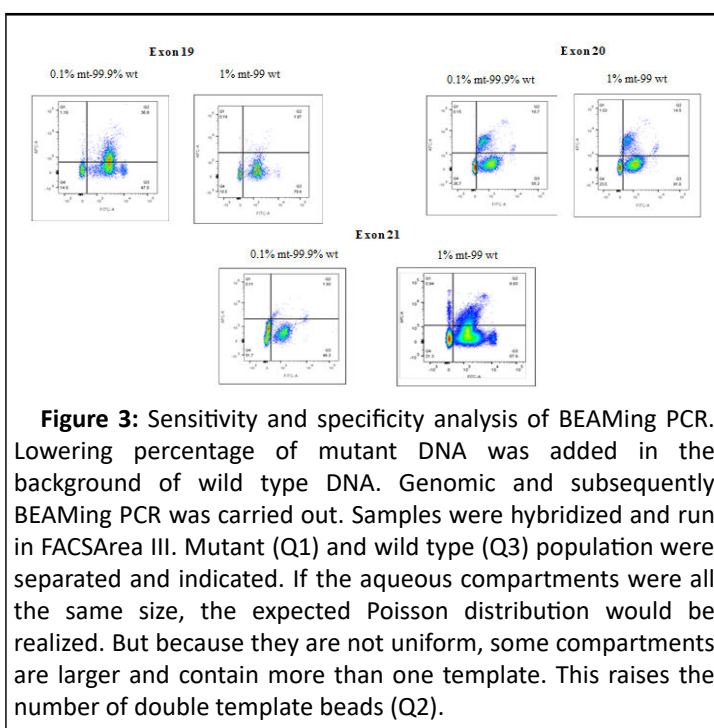
Concentration of ctDNA

We could effectively isolate ctDNA from the plasma of all the samples. The lowest and highest concentration was 5.2 and 31.23 ng/mL with a mean of 15.74 (Figure 1). Moreover, the mean of A260/280 ratio of the samples was 1.81 (Figure 2).



Analysis of accuracy and sensitivity of BEAMing using cell line DNA

We first determined the accuracy and sensitivity of the assay using DNA from PC9 cell line for exon 19 mutation and H1975 cell line for exon 20 and 21 mutations. Each mixture contained a total of 100 ng of DNA, and these mixtures were used as templates for the genomic and subsequently emulsion/BEAMing PCR assay. Wild type *EGFR* DNA from A549 cells were mixed with mutated fragment at 1% and 0.1%. These preparations were subjected to BEAMing PCR. The fractions of the mutated fragment were estimated by the ratio of the numbers of beads labelled with Cy5 (mutant) and those labeled with Alexa 488 (wild-type). For the limit of detection, this assay was able to detect mutant *EGFR* alleles for samples containing as little as 0.1 ng of mutant DNA per 100 ng of DNA. Below this concentration, there was no amplification of the mutant allele (Figure 3).



Mutation assay by BEAMing and its comparison with qPCR data

BEAMing PCR was done on 100 ctDNA samples for the presence of EGFR exon 19, 20 and 21 mutations. Real time PCR was done on tumor tissues for the presence of same mutations. After the hybridization, the samples were run in FACSaria III for the presence of mutations. Percentage true positives and true negatives was calculated with respect to the qPCR data (Therascreen) available in our hospital Electronic Medical Records (EMR) (Tables 3-5) as well as with qPCR data using Easy Line PCR kit (Diatech) (Tables 6-12). BEAMing/EMR-qPCR comparisons were made between 71 (exon 19) (Table 3), 65 (exon 20) (Table 4) and 66 (exon 21) (Table 5) samples respectively. Similarly, for exon 19 and 21, 49, 50 and 49 samples were compared between Diatech qPCR/EMR-qPCR, Diatech qPCR/BEAMing PCR and EMR qPCR/BEAMing PCR respectively (Tables 6-8 and Tables 9-12). Our main aim was to compare our EMR-qPCR results with ctDNA BEAMing assay as the qPCR results along with histopathology results were considered for the eligibility of the targeted therapy of our patients. Our results showed that, for exon 19, 20 and 21 BEAMing data showed high significant concordance when

compared to the EMR-qPCR data. The P value was 0.625 and 1.0. Moreover, when BEAMing data were compared with the Diatech Easy Line qPCR data showed high significance for exon 19 ($P=0.625$, Table 4), exon 20 ($P=1.000$, Table 7) and exon 21 ($P=1.00$, Table 9). Since the p-value is larger than significance level $\alpha=0.05$, we cannot reject null hypothesis and assume that there is no significant difference between BEAMing PCR assay and EMR qPCR and Diatech qPCR assay. Hence, from above results we conclude that there is concordance between BEAMing and EMR qPCR and Diatech qPCR. However, false positives and false negative results were also reported when BEAMing data were compared with EMR qPCR data. High false positives and negatives could be because deamination of nucleotides in FFPE samples which causes the C:G>T:A change, that, in turn, yields a false-positive mutation result. To prevent this, uracil DNA glycosylase (UDG), which cleaves deaminated cytosine (uracil) should be used which was not included in the kit. The results for comparison between exon 20 T790M qPCR and BEAMing and Diatech kit data were not estimable since, all the cases were negative in one of the comparative machines.

Table 3: Exon19_EMR-qPCR vs. Exon19_BEAMing crosstabulation.

			BEAMing PCR		Total
			Negative	Positive	
EMR qPCR	Negative	Count	46	6	52
		% within Exon19_BEAMIng	93.9%	27.3%	73.2%
	Positive	Count	3	16	19
		% within Exon19_BEAMIng	6.1%	72.7%	26.8%
Total		Count	49	22	71
		% within Exon19_BEAMIng	100%	100%	100%

Chi-Square tests

	Value	Exact Sig. (2-sided)
McNemar Test		0.508 ^a
N of Valid Cases	71	

Note: a. Binomial distribution used.

Symmetric measures

		Value	Asymptotic std. error ^a	Approx. T ^b	Approx. Sig.
Measure of agreement	Kappa	0.692	0.095	5.862	0.000
N. of valid cases		71			

Note: a. Not assuming the null hypothesis. b. Using the asymptotic standard error assuming the null hypothesis.

Table 4: Exon20_EMR-qPCR vs. Exon20_BEAMing crosstabulation.

			BEAMing PCR		Total
			Negative	Positive	
EMR qPCR	Negative	Count	63	1	64
		% within Exon20_BEAMIng	100.0%	50.0%	98.5%
	Positive	Count	0	1	1
		% within Exon20_BEAMIng	0.0%	50.0%	1.5%
Total		Count	63	2	65
		% within Exon20_BEAMIng	100.0%	100.0%	100.0%

Chi-Square tests

	Value	Exact Sig. (2-sided)
McNemar Test		1.000 ^a
N of Valid Cases	65	

Note: a. Binomial distribution used.**Symmetric measures**

		Value	Asymptotic std. error ^a	Approx. T ^b	Approx. Sig.
Measure of agreement	Kappa	0.66	0.318	5.656	0.0
N. of valid cases		65			

Note: a. Not assuming the null hypothesis. b. Using the asymptotic standard error assuming the null hypothesis.

Table 5: Exon21_EMR-qPCR vs. Exon21_BEAMing crosstabulation.

			BEAMIng PCR		Total
			Negative	Positive	
EMR qPCR	Negative	Count	59	1	60
		% within Exon21_BEAMIng	96.7%	20.0%	90.9%
	Positive	Count	2	4	6
		% within Exon21_BEAMIng	3.3%	80.0%	9.1%
Total		Count	61	5	66
		% within Exon21_BEAMIng	100.0%	100.0%	100.0%

Chi-Square tests

	Value	Exact Sig. (2-sided)
McNemar Test		1.000 ^a
N of Valid Cases	66	

Note: a. Binomial distribution used.

Symmetric measures

		Value	Asymptotic Std. Error ^a	Approx. T ^b	Approx. Sig.
Measure of agreement	Kappa	0.703	0.162	5.737	0.000
N. of valid cases		66			

Note: a. Not assuming the null hypothesis. b. Using the asymptotic standard error assuming the null hypothesis.

Table 6: Comparison between Exon19_Diatechkit-qPCR vs. Exon19_EMR-qPCR.

Exon19_Diatechkit * Exon19_EMR Crosstabulation		EMR-qPCR			P-Value
		Negative	Positive	Total	
Diatechkit-qPCR	Negative	40 (97.6)	1 (02.4)	41	1.000
	Positive	1 (12.5)	7 (87.5)	8	
	Total	41 (83.7)	8 (16.3)	49	

Interpretation: Since the p-value is larger than significance level $\alpha=0.05$, we cannot reject null hypothesis and assume that there is no significant difference between Diatechkit and EMR qPCR

methods. Hence, from above results we conclude that there is concordance between Diatechkit and EMR qPCR methods.

Table 7: Comparison between Exon19_Diatechkit-qPCR vs. Exon19_BEAMing-qPCR.

Exon19_Diatechkit * Exon19_BEAMing Crosstabulation		BEAMing			P-Value
		Negative	Positive	Total	
Diatechkit-qPCR	Negative	40 (97.6)	1 (02.4)	41	0.625
	Positive	3 (33.3)	6 (66.7)	9	
	Total	43 (86.0)	7 (14.0)	50	

Interpretation: Since the p-value is larger than significance level $\alpha=0.05$, we cannot reject null hypothesis and assume that there is no significant difference between Diatechkit qPCR and

BEAMing methods. Hence, from above results we conclude that there is concordance between Diatechkit and BEAMing methods.

Table 8: Comparison between Exon19_EMR-qPCR and Exon19_BEAMing.

Exon19_EMR * Exon19_BEAMing Crosstabulation		BEAMing			P-Value
		Negative	Positive	Total	
EMR-qPCR	Negative	40 (97.6)	1 (02.4)	41	0.625
	Positive	3 (37.5)	5 (62.5)	8	
	Total	43 (87.8)	6 (12.2)	49	

Interpretation: Since the p-value is larger than significance level $\alpha=0.05$, we cannot reject null hypothesis and assume that there is no significant difference between EMR and BEAMING

machines. Hence, from above results we conclude that there is concordance between EMR and BEAMING machines.

Table 9: Comparison between Exon21R_Diatechkit-qPCR and Exon21R_EMR-qPCR.

Exon21R_Diatechkit * Exon21R_EMR Crosstabulation		EMR-qPCR			P-Value
		Negative	Positive	Total	
Diatechkit-qPCR	Negative	42 (95.5)	2 (04.5)	44	1.000
	Positive	1 (20.0)	4 (80.0)	5	
	Total	43 (87.8)	6 (12.2)	49	

Interpretation: Since the p-value is larger than significance level $\alpha=0.05$, we cannot reject null hypothesis and assume that there is no significant difference between Diatechkit and EMR

machines. Hence, from above results we conclude that there is concordance between Diatechkit and EMR machines.

Table 10: Comparison between Exon21R_Diatechkit and Exon21R_BEAMing.

Exon21R_Diatechkit * Exon21R_BEAMing Crosstabulation		BEAMing			P-Value
		Negative	Positive	Total	
Diatechkit-qPCR	Negative	44 (97.8)	1 (02.2)	45	1.000
	Positive	1 (20.0)	4 (80.0)	5	
	Total	45 (90.0)	5 (10.0)	50	

Interpretation: Since the p-value is larger than significance level $\alpha=0.05$, we cannot reject null hypothesis and assume that there is no significant difference between Diatechkit qPCR and

BEAMing methods. Hence, from above results we conclude that there is concordance between Diatechkit qPCR and BEAMing methods.

Table 11: Comparison between Exon21R_EMR-qPCR and Exon21R_BEAMing.

Exon21R_EMR * Exon21R_BEAMing Crosstabulation		BEAMing			P-Value
		Negative	Positive	Total	
EMR-qPCR	Negative	42 (97.7)	1 (02.3)	43	1.000
	Positive	2 (33.3)	4 (66.7)	6	
	Total	44 (89.8)	5 (10.2)	49	

Interpretation: Since the p-value is larger than significance level $\alpha=0.05$, we cannot reject null hypothesis and assume that there is no significant difference between EMR qPCR and

BEAMing methods. Hence, from above results we conclude that there is concordance between EMR qPCR and BEAMing methods.

Table 12: Comparison between Exon21Q EMR-qPCR and Exon21Q_BEAMing.

Exon21Q_EMR * Exon21Q_BEAMing Crosstabulation		BEAMing			P-Value
		Negative	Positive	Total	
EMR-qPCR	Negative	47 (100.0)	0 (00.0)	47	1.000
	Positive	1 (50.0)	1 (50.0)	2	
	Total	48 (98.0)	1 (02.0)	49	

Interpretation: Since the p-value is larger than significance level $\alpha=0.05$, we cannot reject null hypothesis and assume that there is no significant difference between EMR qPCR and BEAMing methods. Hence, from above results we conclude that there is concordance between EMR qPCR and BEAMing methods.

Discussion

Circulating tumor DNA was studied for the analysis of *EGFR* mutations in NSCLC patients in a number of studies. We have reported here that ctDNA is a promising biomarker for the early detection of *EGFR* mutations as well as following the course of therapy in NSCLC patients. We could effectively isolate and measure the ctDNA concentration from as many as 100 patient samples. However, whether ctDNA levels are exactly proportional to the tumor burden cannot be ascertained conclusively because varying level and because there is no independent way to measure systemic total burden at this time

(Figures 1 and 2). Many studies have shown the sensitivity of detecting gene mutation in plasma ctDNA is largely dependent on detection techniques. Among many techniques, BEAMing PCR was able to detect a very small amount of mutant DNA sequences in a larger pool of fragments containing wild-type DNA, in order of a single mutant allele in a background of 10,000 wild-type alleles, and it is able to enabling copy-number quantification [22,23]. The technique is based on a combination of emulsion digital PCR and flow cytometry, with beads, emulsification, amplification and magnetics to achieve the necessary level of sensitivity.

In the present study we were comparing the detection concordance between tissue *EGFR* qPCR techniques (both Therascreen and Diatech kit) with that of ctDNA BEAMing PCR. We had used the primer sets designed by Diehl et al. [21] which could able to detect 8 different exon 19 mutations in

addition to T790M exon 20 and L858R and L861Q exon 21 mutations. All the wild type primers were labeled with Alexa Fluor 488 probe and all the mutant probes were labeled with Cy5 probe (Table 2). As the fraction of mutant alleles in the isolated DNA were minimal, we first designed the sensitivity assay by mixing different percentages of mutant DNA in the background of wild type DNA to see whether BEAMing PCR could detect the mutant alleles. Our results showed that this assay could detect mutant alleles as low as 0.1% in the background of 99.9% of wild type allele. BEAMing PCR was done in all the samples. But BEAMing PCR comparisons with EMR-qPCR (Therascreen) and Diatech qPCR could be done with lesser number of samples because of non-availability of all the qPCR data (Tables 3-12). Concordance percentage was high when BEAMing data were compared with qPCR data. Since the p-value is larger than significance level $\alpha=0.05$, null hypothesis cannot be rejected and we can assume that there is no significant difference between EMR and BEAMing methodology. Hence, from above results we conclude that there is concordance between EMR qPCR/Diatech qPCR and BEAMing methods.

This study has several limitations. This was a single-institution study where sample size was small. Repeat biopsies were not performed where tissues were inadequate to do the qPCR. However, based on the positive and meaningful results in this study, we can conclusively say that BEAMing PCR is an effective and sensitive assay to know the mutation status of the EGFR gene in NSCLC patients. However, large sample studies involving thousands of patients in a multicentric setup is required to effectively determine the mutation. This will further validate the conclusion of the present study

Conclusion

In summary, although clinical and radiographic characteristics associated with different histology subtypes of NSCLC have been identified, histology alone is unlikely to remain as the primary determinant in the selection of appropriate treatment. Results from a pilot study in breast cancer suggest that blood-based mutation analysis might be used to track patients' response to therapy. This technique would provide an alternative to invasive biopsies that retrieve tumor tissue for analysis. Moreover, future treatment decisions for lung cancer are likely to be based on molecular subtypes reflecting tumor biology rather than clinical features or histologic subtypes. Discoveries linking two areas of seemingly unconnected research tend to reframe existing knowledge in ways that permit new interpretations and new understanding of the field of study addressed by this research.

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Conflict of Interest

The authors declare that there is no conflict of interest.

Author's Contribution

All authors read and approved the final manuscript.

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